EXPLAINABLE AND ROBUST DEEP LEARNING FOR ANOMALY DETECTION INDUSTRIAL CONTROL SYSTEMS

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Introduction

- Industrial Control Systems (ICS) operate critical infrastructure such as water treatment plants.
- Sensors monitor physical processes, while actuators control operations.
- Cyberattacks can manipulate sensor readings and actuator behaviour.
- Deep learning can identify anomalies by learning patterns of normal operation.

Focus: Accurate, explainable and noise-robust anomaly detection.

Problem Statement and Objectives

Problem Statement

- Traditional anomaly detection systems may generate excessive false alarms due to changing operating conditions and sensor noise.

Objectives

- Implement a DAICS-inspired WDNN baseline.
- Evaluate GRU as an alternative temporal prediction model.
- Identify sensors contributing to anomaly predictions.
- Compare detection performance under clean and noisy conditions.

Literature Review and Research Gap

Existing Approaches

- DAICS (Abdelaty et al.): WDNN-based sensor forecasting with adaptive threshold tuning.
- RNN-based methods: Model temporal dependencies in industrial sensor data.
- CNN-based methods: Extract local time-series patterns for anomaly detection.

Research Gap

- Limited direct comparison of WDNN and GRU under matched evaluation conditions.
- Need for interpretable sensor-level anomaly explanations.
- Need to examine performance stability under sensor noise.

Our Contribution: Comparative evaluation of WDNN and GRU with explainability a robustness analysis.

Proposed Methodology

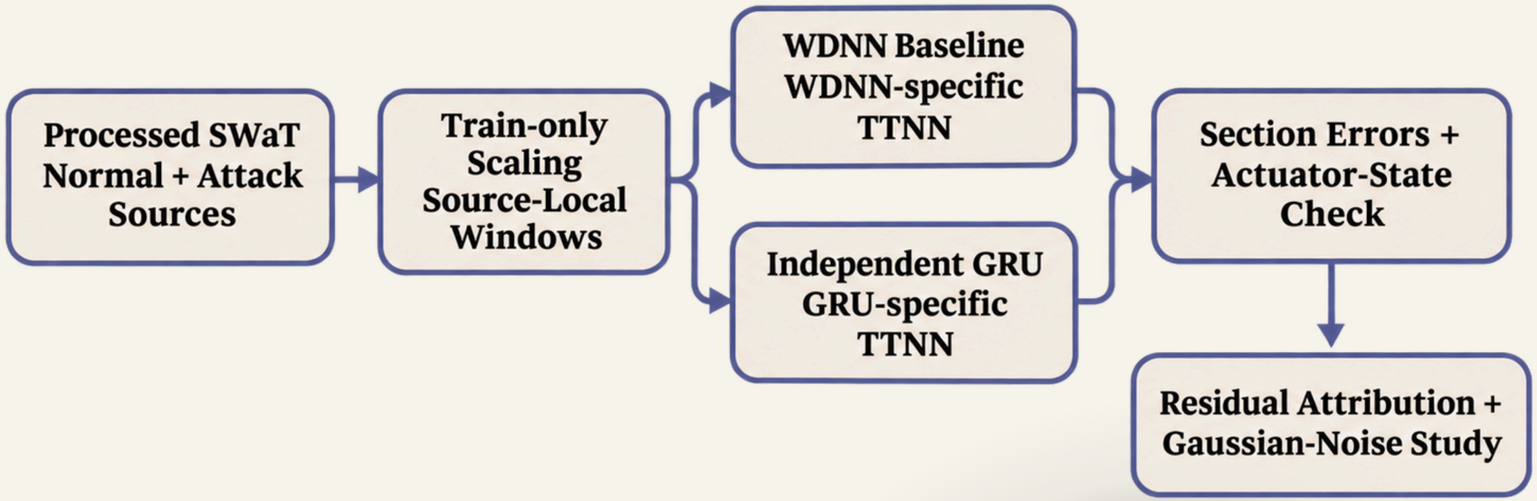
Overall Workflow

SWaT Dataset → Preprocessing → WDNN / GRU → TTNN Threshold Calibration → Anomaly Detection

Additional Evaluation

- Residual-based sensor attribution for explainability.
- Controlled Gaussian noise for robustness testing.
- Direct WDNN versus GRU performance comparison.

Both models are trained and calibrated independently using the same data conventions.



Dataset and Preprocessing

Dataset: Secure Water Treatment (SWaT)

- 44 retained process variables: 24 sensors and 20 actuators.
- Removed 495,000 duplicated normal records.
- Excluded seven unavailable process variables.
- Applied training-fitted MinMax normalization to sensor readings.
- Used normal-only training and validation with separate test sources.

Final Dataset

- Normal: 892,098 records
- Attack: 54,621 records

Model Architectures
WDNN vs GRU

WDNN Baseline

- Wide branch and deep convolutional branch.
- Learns feature relationships and temporal patterns.
- Three sensor prediction sections.

GRU Alternative

- Single GRU layer with 64 hidden units.
- Models sequential dependencies in sensor and actuator data.
- Three independent sensor prediction heads.

Common Configuration

- Input: 44 features × 60 time steps.
- Output: Four future steps for 24 sensors.

Experimental Setup

Environment: Python, PyTorch, NVIDIA RTX 3050 (6 GB).

Detection Configuration

- Input window: 60 samples.
- Forecast horizon: 50 samples.
- Prediction window: 4 samples.
- Anomaly persistence: 30 consecutive samples.

Metrics: Precision, Recall, F1-score and False Positive Rate.

Experimental Setup

Parameter	WDNN	GRU
Optimizer	SGD	Adam
Learning Rate	0.01	0.001
Batch Size	32	32
Epochs Run	100	15
Evaluated Epoch	100	12

Clean Detection Results

Performance Results

Metric	WDNN	GRU
Precision	0.5153	**0.6675**
Recall	**0.8818**	0.8261
F1-Score	0.6505	**0.7383**
False Positive Rate	15.38%	**7.63%**

Key Findings

- GRU achieves higher pooled F1-score and fewer false positives.
- WDNN retains higher attack detection recall.
- The models exhibit different precision–recall trade-offs.

Explainability and Robustness

Explainability

- Used squared prediction residuals to rank contributing sensors.
- Reported the top five sensors for each detected section episode.
- Achieved 96.6% rank-one agreement across 148 matched WDNN-GRU episodes.

Robustness

- Added Gaussian sensor noise at low, medium and high levels.
- Reused identical noise realizations for both models.
- GRU maintained higher pooled F1-score across all noise conditions.

High-Noise F1: WDNN = 0.6557 | GRU = 0.7554

Residual attribution measures error contribution, not causality.

Conclusion and Future Scope

Conclusion

- Developed and evaluated WDNN and GRU-based industrial anomaly detectors.
- GRU achieved higher pooled F1 and lower false-positive rates.
- WDNN retained higher attack detection recall.
- Residual attribution provided interpretable sensor-error rankings.
- GRU maintained its advantage across the tested noise conditions.

Future Scope

- Online TTNN updates and model adaptation.
- Attention-based temporal modelling.
- Evaluation across individual attack intervals and additional datasets.

THANK YOU